



AKTU B.Tech 1st Semester



Engg. Physics

Unit-5 : Nano Materials

ONE SHOT Revision

**Pdf Notes
PYQs
DPP**



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Unit-5: Superconductors and Nano-Materials

Superconductors

7 + 7

- Temperature dependence of resistivity in superconducting materials
- Meissner effect
- Temperature dependence of critical field, Persistent current
- Type I and Type II superconductors
- High temperature superconductors
- Properties and Applications of Super-conductors.

Nano-Materials

7 + 2

- Introduction and properties of Nano materials
- Basics concept of Quantum Dots, Quantum wires and Quantum well
- Fabrication of Nano materials -Top Down approach (CVD) and Bottom-Up approach (Sol Gel)
- Properties and Application of Nano materials.

- Nano is a Greek word means dwarf

1 nm = one-billionth of a meter ($1\text{nm} = 10^{-9}\text{m}$)

Nano size OR Nano scale (1 nm-100 nm)

- Nano materials are those materials which have at least one dimension in Nanometer scale (Between 1nm to 100 nm).

Nano materials can be metal, ceramics, polymers, Composites etc.

- At this scale, the materials exhibit unique physical, chemical and biological properties that are significantly different from those of their bulk counterparts.
- These properties arises due to the increased surface area to volume ratio, quantum confinement effects and other phenomena that occur at the Nano scale.

Surface
volume

- Quantum confinement effect occurs when the dimensions of a material become small enough (nanoscale size) that the motion of charge carriers (electrons or holes) is restricted in one or more spatial directions
- Quantum confinement effect fundamentally alters the electronic and optical properties of materials at nanoscale

Nano Science

- Nano Science is the study of matter and their properties at the Nanometer scale (Between 1nm to 100nm)

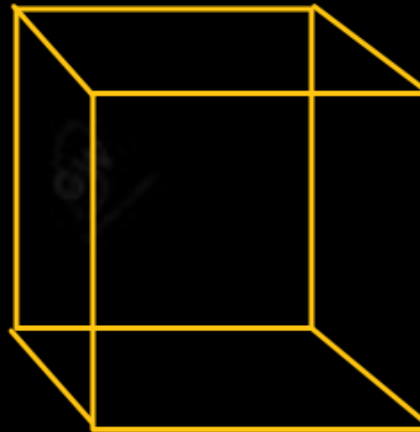
Nanotechnology

- Nanotechnology is the application of Nano science principles to design, synthesize and manipulate materials, devices and systems at the nanoscale. (Between 1nm to 100nm)

Quantum Well (Two dimensional structure)

- If one dimension is reduced to the Nanometer range while other dimension remain same, then the structure formed is known as Quantum well.
- A quantum well is a thin, two dimensional structure in which charge carriers (electrons or holes) are confined in one dimension while they can move freely in other two dimensions.
- Quantum well are widely used in various advanced technologies, including lasers, LEDs, high speed transistors and quantum computing devices

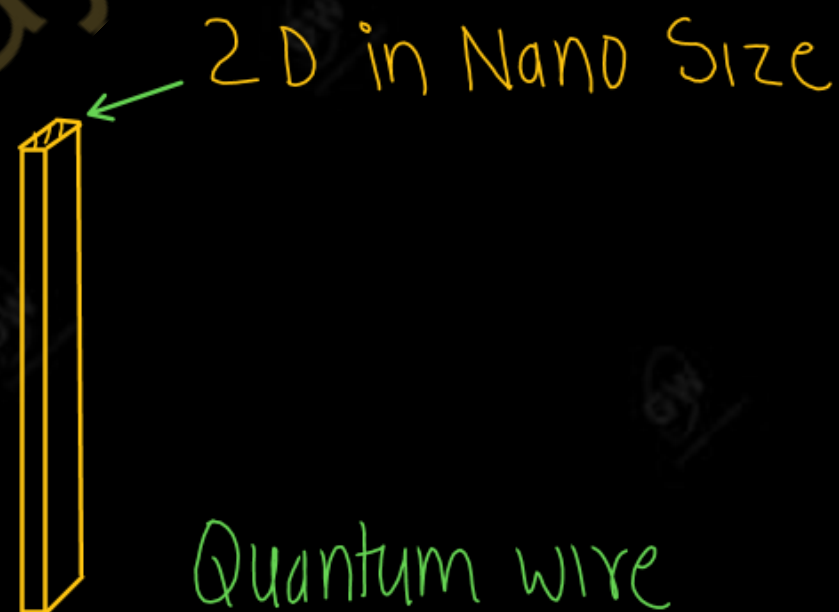
Bulk



Quantum
Well

1D in Nano Size

- If two dimension are reduced to Nanometer range while the third dimension remain the same, then the structure so formed is called Quantum wire.
- A quantum wire is a nanoscale structure in which charge carriers (like electrons or holes) are confined in two dimensions, allowing free movement only in one dimension.
- Quantum wire are used in quantum computing, photonics, energy technologies etc. Their size and structure make them highly adaptable for future technologies.



Quantum Dots (Zero dimensional structure)

- When all the three dimensions of the material are reduced to Nanometer range, then it is called as quantum Dot. They are zero-dimensional nanostructures.
- Quantum dots are nanoscale semiconductor particles that exhibit unique electronic and optical properties due to quantum confinement effects.
- Quantum dots confine charge carriers (like electrons and holes) in all three spatial dimensions.
- Quantum dots have a wide range of applications in optoelectronics, biological imaging, photonic devices, quantum computing and environmental sensing



3D in nm Size

Quantum Dot

- There are two main approaches for the Fabrication or Synthesis of Nano materials.

(i) Top down Approach

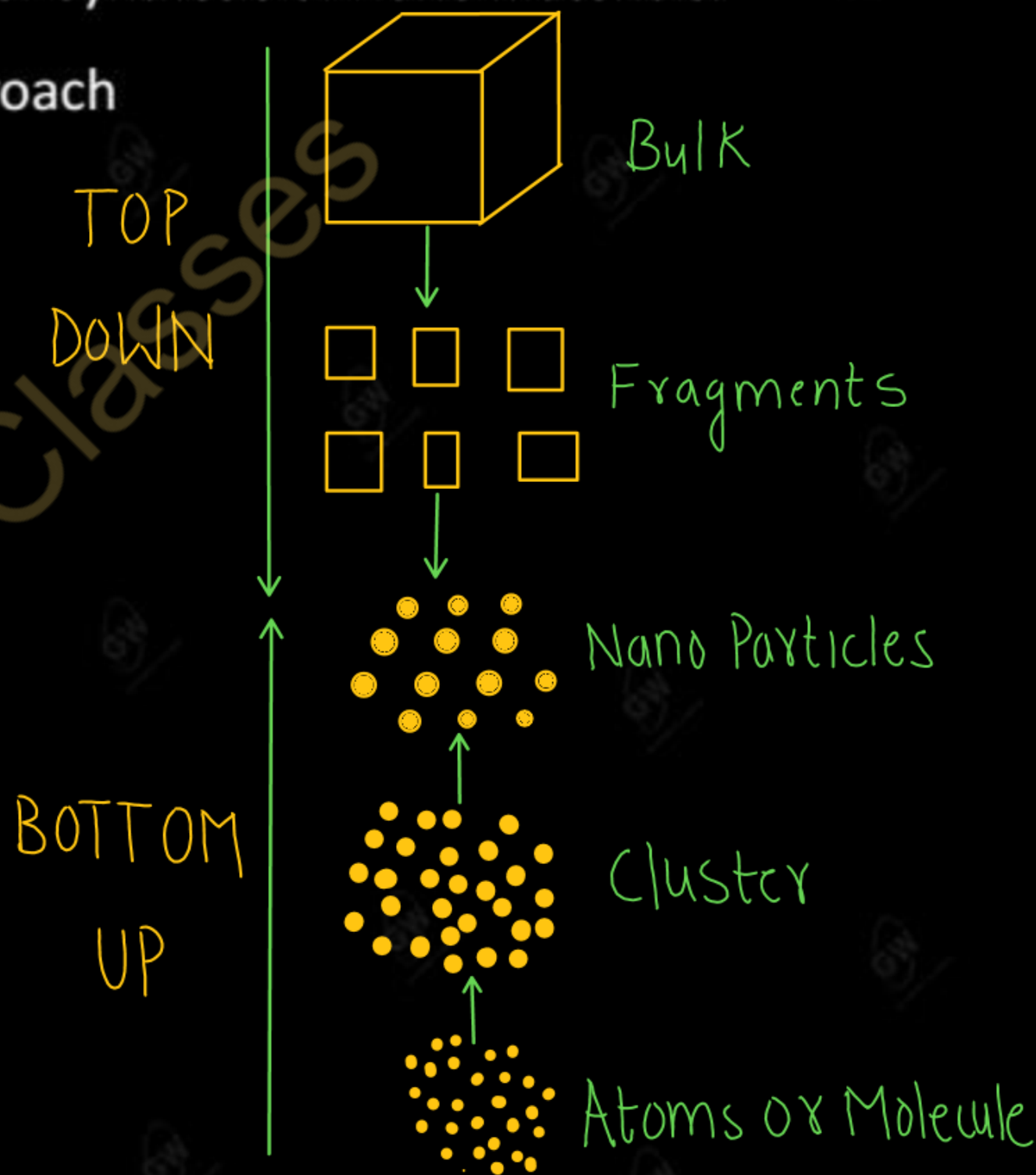
(ii) Bottom-up Approach

Top down Approach

- Top down Approach involves breaking down bulk materials into Nano scale Structures using various physical, chemical and mechanical process.

Bottom-up Approach

- Bottom-up Approach involve building Nano materials from the atomic or molecular level, essentially assembling them atom by atom or molecule by molecule.

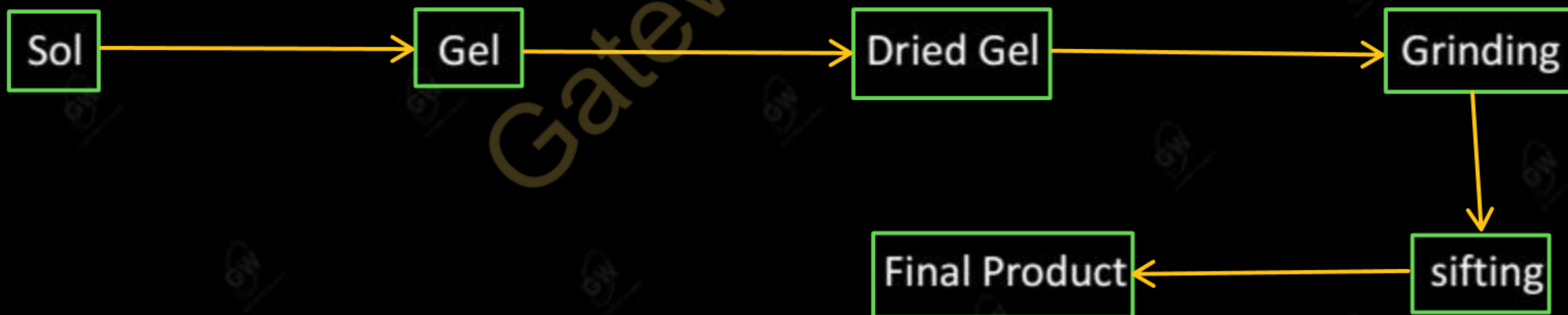


- Sol-Gel method
- Chemical vapor deposition method

Sol-Gel Method

- Sol-Gel method is a powerful and versatile technique for producing nanomaterials with precise control over their size, morphology and composition.
- It is a chemical process to synthesize nanomaterials.

Steps involved in Sol-Gel Method



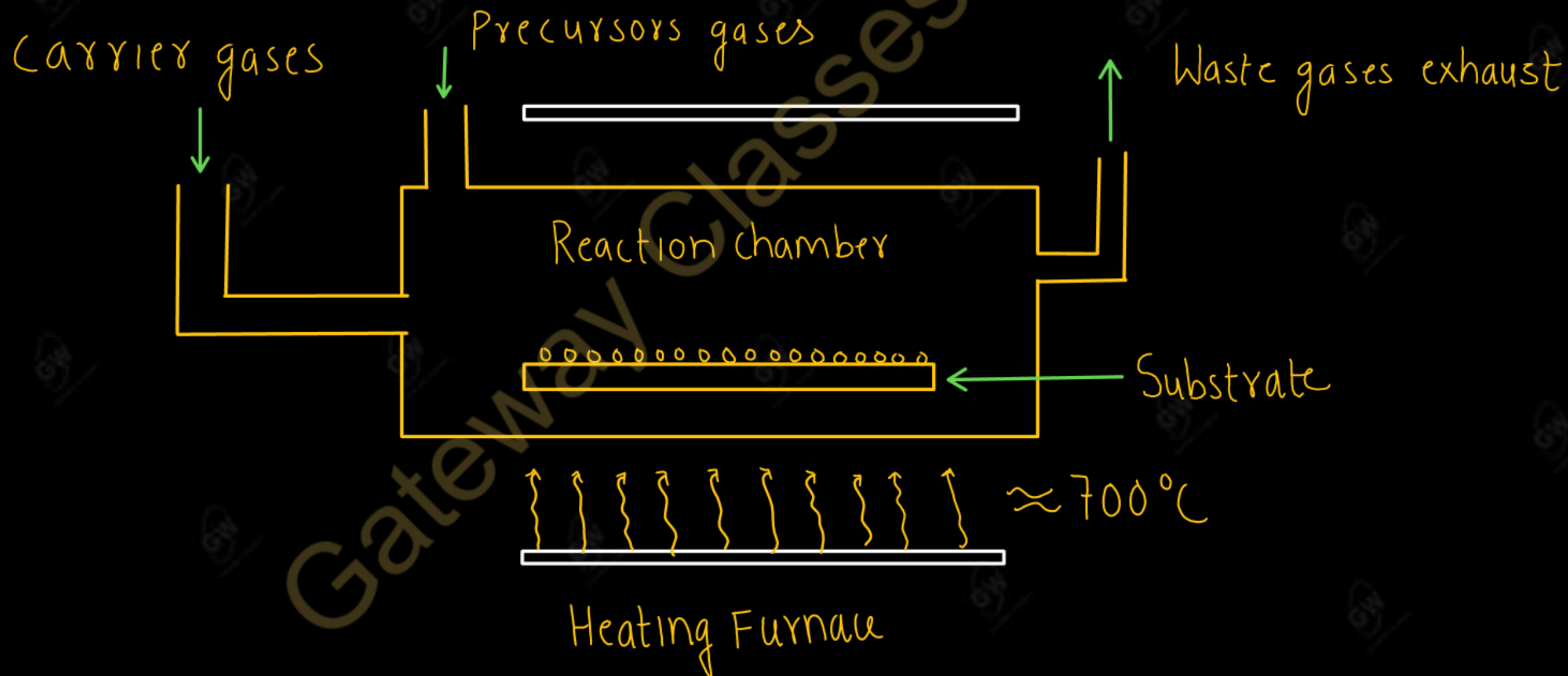
1. The sol is a colloidal suspension of nanoparticles (the dispersed phase in which size of particles is so small that gravitational forces do not exist)
2. A gel is a semi-rigid mass that forms when the solvent from the Sol begins to evaporate and the particles left behind begin to join together in a continuous network.
3. The remaining liquid is removed using drying process accomplished by thermal treatment and remains left as dried gel.
4. The dried gel is grinded to get the material in to powdered form.
5. Once the powder is obtained its lumps are removed using shifting process
6. The final product can be deposited on a substrate to get a thin film or we can cast it into desired shape.

Advantages of sol-gel Method

- (i) It is a cheap and low temperature technique
- (ii) Fine control over composition and structure
- (iii) Rate of reaction can be easily controlled
- (iv) High chemical purity

Gateway Classes

- Chemical Vapor deposition (CVD) is a widely used method for producing nanomaterials with precise control over their composition, structures and properties.



- CVD process take place in a reaction chamber that is heated to a high temperature.

- The process begins by introducing one or more gaseous precursors compounds in to a reaction chamber.
- The precursor gases undergo chemical reaction when exposed to heat in reaction chamber
- These reactions breakdown the precursor gases, causing the desired material to deposit on to a substrate
- This deposition forms final nanomaterials
- Any unwanted byproducts from the chemical reaction are carried away by a flow of carrier gas
- The thickness and properties of the film can be controlled by adjusting the temperature, pressure, gas flow rate of the precursors as well as the duration of the deposition process.
- Nanomaterials produced by CVD are Carbon nanotubes(CNTs), Graphene, nanowires, Quantum Dots, thin film

Advantages of CVD

- It is used to produce high purity nanomaterials
- Excellent control over film thickness and composition
- Capable of producing a wide range of nanomaterials.
- Produce uniform thin films

Nanomaterials possess unique properties compared to their bulk counterparts due to their small size, high surface area to volume ratio, quantum confinement effects and other phenomena that occur at the Nano scale.

(1) High surface area to volume ratio

(2) Mechanical Properties

- Nano materials, such as carbon nanotubes and graphene, can exhibit significantly higher strength compared to bulk materials
- Some nanomaterials, such as nanowires and nanotubes possess superior flexibility and toughness

(3) Magnetic Properties : The smaller particles are more magnetic than bulk material.

(4) Electrical Properties : Nano materials can exhibit a wide range of electrical conductivities

(5) Optical Properties Nanomaterials can show unique optical properties, such as enhanced absorption, scattering and fluorescence.

(6) Thermal Properties : Some nanomaterials have higher thermal conductivity than bulk materials.

(7) Highly Reactive : Nanomaterials are chemically more reactive than their bulk counter parts.

Materials	Macro scale	Nano scale
Copper	opaque	Transparent
Platinum	Inert	Catalytic
Aluminium	Stable	Combustible
Silicon	Insulator	Conductor
Gold	Solid at room temperature	Liquid at room temperature

Nanomaterials have a wide range of applications across various fields

(1) Electronics : Nano materials, especially carbon nanotubes and quantum dots, are used in the development of smallest, faster and more efficient electronic devices.

(2) Medicine

(i) Drug Delivery: Nanomaterials can be used to deliver drugs more precisely to targeted areas in the body, particularly in cancer treatment.

(ii) Diagnostics: Nanotechnology is helpful in medical diagnostics by providing faster, cheaper and portable diagnostics equipment's like MRI and CT scans for better visualization of tissue and organs.

(i) Solar Cells : Nano materials like quantum dots and quantum wires are used to improve the efficiency of solar cells.

(ii) Batteries: Nano materials are used to improve the performance of batteries by improving energy storage, capacity and charging speed (such as lithium-ion batteries)

(4) Material Science

Nano materials are used to develop stronger and lighter materials for use in aerospace, automotive and construction industries

(5) Environmental protection

Nanomaterials are used for water purification and Air purification

(6) Agriculture

Nanomaterials are used in the development of pesticide and fertilizers

There are several disadvantages of nanomaterials

Health Risks

- Due to their small size and high reactivity, nanomaterials can penetrate cell membranes and interact with cellular components, potentially causing damage. This could lead to inflammation, oxidative stress, and cell death.
- Nanoparticles, due to their small size, can be inhaled deep into the lungs, where they may cause irritation or inflammation
- Nanoparticles may penetrate the skin more easily than larger particles, causing potential irritation or allergic reactions.

Environmental Impact:

- Their small size and reactivity could make them more difficult to remove from ecosystems, potentially contaminating soil and water

UNIT - 5 : DPP - 2

TOPIC : Nano-Materials

Q.1 What is top down and bottom up method. (2)

AKTU 2024 (Sem-II)

Q.2 What are the changes in the properties that take place in a material when its size reduced to nanoscale.

Give some examples. Also explain the reasons for these property changes. (7)

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2024(Sem-II)

Q.3 Explain quantum confinement effect in nanomaterials? (2)

AKTU 2024(Sem-I)

Q.4 What is the purpose of Nanoscience? Discuss any one method (CVD/Sol-Gel) for the synthesis of nanomaterials. (7)

AKTU 2024(Sem-I)

Q.4 What do you mean by a Quantum Well? (2)

AKTU 2023 (Sem-II)

Q.5 Discuss properties and the potential applications of nano materials. What are the risks of using nano materials to human body? (7)

AKTU 2023 (Sem-II)

Q.6 Write down any two applications of the Nano materials. (2)

AKTU 2023 (Sem-I)

Q.7 What are Nano materials? Explain briefly the basic concepts of Quantum Dots, Quantum wires and Quantum well. (7)

AKTU 2023 (Sem-I)

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Thank You